

A reference architecture for digital twin-based supply chain information exchange

The PURIS case

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Abstract: In this design science research project, based on the Catena-X automotive data space, we built a novel reference architecture for digital twin-based supply chain information sharing. The artifact aims at increasing supply reliability by sharing short-term information that is commonly not shared within a network of actors due to its sensitivity. Our architecture addresses these issues by providing measures for sensitive and compliant information sharing. Besides data sovereignty and compliance, we derived and fulfilled the design objectives of interoperability, up-to-date information, flexible information sharing and adoption. Our design project ran through two major design iterations of building and evaluating the architecture through the instantiation of a reference implementation that has been published as free open-source software.

Keywords: Data Ecosystem, Data Space, Digital Twin, Supply Chain Resiliency

1 Introduction

Supply chain resiliency is the ability to be prepared and handle disruptive events more efficiently to gain competitive advantages [Ho15]. One prominent incident raising awareness for supply chain resiliency is the container ship “Ever Given” blocking the Suez Canal in 2021, which resulted in a cost of \$88.79m [Tr25]. To improve supply chain resiliency, collaboration and information sharing are considered key elements [Ho15]. Information sharing is the process of providing access to information to others [Ju23], and information exchange is the “technical conversion process of [information]” and a subprocess of information sharing [Ju23]. Traditional information sharing within supply chains often covers order information, while broader sharing scopes capture additional

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information, such as inventory [CF00]. During shortages, the information exchange is conducted via non-standardized combinations of mail, phone, and spreadsheets. During operations, Electronic Data Interchange (EDI) is used as a well-established technology in supply chains [GH88]. Existing EDIFACT standards provide capabilities to manage economic and logistics transactions, such as demand forecasts, orders, invoices, and deliveries [VD19]. Discussing the application of EDI during shortages with practitioners reveals several gaps: First, the defined message formats do not provide information on the detail level needed. Second, EDIFACT has been designed for efficient information exchange that happens daily and covers all products within the customer-supplier-relationship. The information exchange increases the time to action by hours due to the processing times on the sender and receiver sides. Third, EDI does not provide mechanisms to preserve data sovereignty. Additional information exchanged during bottlenecks is sensitive. Participants fear that the information could be used against them. Data Ecosystems may be applied to foster supply chain resiliency using information exchange. These are networks of socio-technical actors engaging in beneficial relationships based on data as the transaction object [OL18], [OBF19]. Similarly, Industry 4.0 technologies improve the flexibility of production and supply chain processes [IDS19]. Others have analyzed the benefits of applying Industry 4.0 technologies to supply chain resiliency, such as [SB21], [KGK22]. More recently, digital twins are used for integration and simulation purposes along business processes [Iv19], [WW19].

Because of the above, our research objective is the following: *Design a reference architecture for digital twin-based supply chain information exchange, preserving trust for sensitive information.*

We followed the design science research (DSR) paradigm to build and evaluate a novel artifact in the form of a reference architecture for information exchange. We performed two main design iterations within a large consortium over about two years. Doing so, we could ensure a close collaboration between scholars and practitioners, which helped us to better understand the actual problem space and build solutions to fulfill real-world needs. Our work contributes to improving interoperability in sharing short-term supply chain information. Further, our architecture and information definition allow us to share the latest information that may also be more independent of order variance and thus may contribute to reducing the bullwhip effect [LPW97].

The paper is structured as follows: After the Introduction, we outline the background of our paper (Section 2). Then, we elaborate on our research design (Section 3). Following, we explain the design objectives for the architecture design and the final reference architecture (Section 4). Next, we explain how we instantiated the reference architecture (Section 5). Finally, we discuss whether and how the design objectives have been applied, including limitations of the approach and the artifacts created (Section 6).

2 Research background

2.1 Information sharing in supply chains

Within supply chain management material, information and financial flows are managed to source, produce and deliver goods in a network of participants [Ta06]. Companies in this network often operate based on a pull principle, meaning that the supply chain activities are triggered by customer orders [We20]. Information such as customer and supplier orders are part of the information flow. This transactional information is exchanged electronically via standardized formats such as EDIFACT [GH88]. While most authors refer to any information exchanged in supply chains as “information sharing” [Sh01], some authors only consider information additional to the minimally needed transactional information (orders) as *information sharing* [GRR14]. Information sharing receives high interest in the domain of supply chain resiliency [CP04], [Ho15], [Sh01]. Supply chain resiliency refers to a supply chain’s ability to respond to disruptions and recover from these to gain competitive advantages [Ho15]. Information sharing is considered the basis for collaboration [CP04]. It improves supply chain visibility [JM11] that can mitigate the bullwhip effect by allowing better handling of variance in orders [LPW97]. Recent global disruptions, such as COVID, have increased the overall interest of companies in higher visibility and integration. data ecosystems like Catena-X provide an infrastructure that can be reused for multiple use cases to lower costs. Catena-X uses a data space as the technical infrastructure [LH24].

2.2 Data ecosystems and data spaces

Data ecosystems are „socio-technical complex networks in which actors interact and collaborate to find, archive, publish, consume, or reuse data as well as to foster innovation, create value, and support new businesses.“ [OBF19] Besides the economic benefits emphasized in this definition, further benefits can be expected from improvements in 1) socio-political aspects, 2) easier production and consumption of data, 3) communication and interaction between actors, and 4) improvements in the data and service quality [OBF19]. Data ecosystems consider data in its different forms as an asset and a key transaction object. These forms vary from the information represented by the data to services or applications manipulating and using data to fulfill requirements [Mö24]. Data ecosystems benefit from data sovereignty mechanisms as incorporated by data spaces to share sensitive information selectively [LH24]. A data space is a data sharing infrastructure used to implement data ecosystems technically. Data spaces are primarily used for inter-organizational information exchange [Mö24]. To consume data, peer to peer communication between a data provider and a data consumer is used. Data and their usage conditions are discovered via a provider’s catalog directly or via a federator. Within this process, only meta information is exchanged. The asset itself can only be consumed after the partners have contracted the asset usage [Ot22a].

3 Research design

To design our reference architecture, we followed a *problem-solving DSR configuration* [Sc24] because the close collaboration with practitioners allows us to get a deep understanding of the problem space. The DSR project was part of a consortium of 28 companies in Germany. The core team within the DSR project consisted of one automotive OEM, one research organization, and two factory outfitting companies. While the OEM provided insights on big companies, the factory outfitters provided insights on small and medium-sized enterprises that cover many of their customers. Within the period of August 2022 to July 2024, the project team had met in more than 340 appointments, including eight internal and five external evaluations. The appointments have been used to align on asynchronous working packages, work collaboratively, and provide reviews. Internal evaluations cover demonstrations of the last three-month iteration’s progress, while external evaluations capture workshops with external representatives to gather feedback on the information exchanged. Two major design iterations across two years have been conducted to fulfill our goal of designing a novel artifact for data sharing (see Fig. 1). Design Iteration 1 clarified the scope from a business-driven point of view in terms of information that needs to be exchanged. From an architecture perspective, it has been used to integrate the data space-related components. Design Iteration 2 integrated digital twins into the architecture. The final architecture will be explained in Section 4.2.

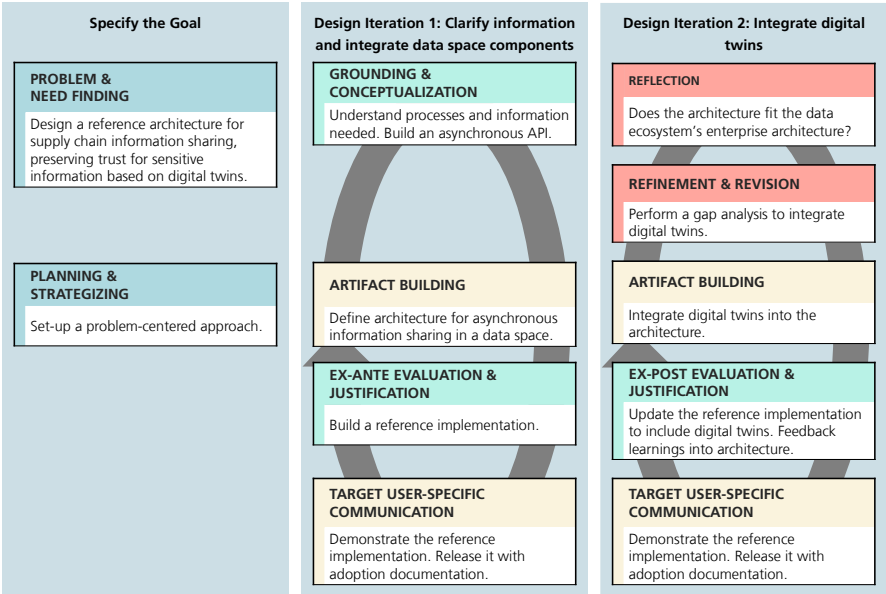


Fig. 1 Overview of design iterations presented with BAUSTEIN

In **Design Iteration 1**, we started by *grounding* our artifact. A basic understanding of

supply chain information exchange has been built based on the practitioners' knowledge captured within the team. The main parts of the knowledge from Oprel et al. [Op24] have been reused. While defining the information set needed, the team considered information provisioning. During the *design* of the architecture, we derived and formulated a set of design objectives (see Section 4.1). Besides reusing knowledge, the derivation process has been conducted by abstracting from work package results and constraints from within consortia or the law. Tab. 2 provides further information on the single design objectives' origins. Initially, an asynchronous API has been designed that is exposed to partners under specific conditions via data space components. To *evaluate* the architecture, a reference implementation has been created. A factory outfitter has implemented the provider side to show data ingestion based on a machine execution system. The information exchange has been presented in a system demonstration within a SAFE organization using the two applications and the data space infrastructure. The reference implementation has been released as free open-source software (FOSS). Besides the software release, additional documentation has been provided elaborating on the business side and how to operate the software for adoption. The reference architecture has been published in the form of normative documentation tailored for the defined information set.

In **Design Iteration 2**, we *reflected* on whether our architecture fits the ecosystem approach designed within the consortia by other teams. As the ecosystem merely relies on digital twins, a gap analysis has been performed to identify the delta and define the needed architectural work to integrate digital twins. This analysis has been performed considering consultation from the digital twin-related teams within consortia and relying on those teams' documentation. The identified architectural work has been conducted to *refine* the reference architecture. The reference implementation has been updated accordingly while providing feedback within the team and the consortium on implementing the digital twins. The reference implementation has been *evaluated* again during a system demonstration. In this iteration, the reference implementation has been used for both the data consumer and the data provider side, while using the data space infrastructure and exchanging information based on digital twins. Finally, for *communication* purposes, the reference implementation and the documentation have been released in an updated version.

4 Designing a reference architecture for digital twin-based supply chain information exchange in data spaces

4.1 Deriving design objectives

To derive a purposeful artifact, we start by formulating design objectives (DOs). These objectives cover the key underlying requirements derived from empirical data collected during the engagement with industry and literature, as listed in the Appendix.

DO1 – Compliance by design: All artifacts created within the project must comply with

German antitrust law (“Kartellrecht”) and the German act against restraints of competition (“Gesetz gegen Wettbewerbsbeschränkungen”). Thus, third-party information must not be revealed, and the information exchange must not restrict or distort competition.

DO2 – Up-to-date information: Information in EDIFACT standards covers order-level transactions and is exchanged once a day. Processing large files is time-consuming and increases the time to action. When evaluating and preventing supply shortages, partners assess information multiple times a day. Seeing the latest available information in these cases improves decision-making.

DO3 – Interoperability: Interoperability is a key for data ecosystems, especially of data spaces as an information exchange architecture [Ot22a]. Interoperability can be addressed on different levels [Eu17] and allows to switch software, if all software solutions comply with the same interface definitions and associated behavior. Hence, software solutions must follow the same semantics and protocols to enable the same business capabilities.

DO4 – Flexible information exchange on a per-material level: EDIFACT covers all materials in a relationship at once. This increases the processing times of interface messages. To increase flexibility, information must be exchanged per material (*DO 4.1*). Thus, the information exchange can take place, for instance, for only a specific control unit. Further, only exchanging relevant information allows the provider to stay in control. Due to multiple domains involved, it should be easy to add further information (*DO 4.2*).

DO5 – Multi-use information: Supply chains are commonly structured hierarchically or polycentrically. Automotive supply networks are commonly considered focal but contain polycentric structures for generic components used in multiple supply chains [We20]. To allow reuse and standardization by scaling along the supply chain, the information exchanged must be domain unspecific (*DO 5.1*). To increase the value of the information exchanged, the information must apply to multiple use cases within the supply chain domain (*DO 5.2*), such as resolving shortages and monitoring the fulfillment.

DO6 – Scaled architecture usage: Supply chain-related information exchange scales along the partners exchanging information. A reference implementation should be developed (*DO 6.1*) to promote adoption by easing the understanding and providing a basis for participation. Similarly, the documentation for adoption must be published using a permissive license as free open-source (software) (*DO 6.2*). Following the argumentation of DO3, software components should be interchangeable (*DO 6.3*).

DO7 – Data sovereignty: Current supply chain information exchanged commonly reflects the minimally transactional information via e.g., EDIFACT. To allow advanced activities, such as shortage detection and supply chain monitoring, additional information must be exchanged, including delivery forecasts, stock levels, production plans, and material consumption. Exchanging this sensitive data requires trust. Data sovereignty can be understood as “[c]oncepts of self-determination and the capability of a data provider to keep control over their own data.” [Hv23] Thus, the architecture must allow data providers to preserve data sovereignty.

4.2 Building a reference architecture

The component diagram in Fig. 2 illustrates the final version of our architecture (see Appendix for the first version). To increase readability, the components are reported participant-wise, excluding identity and access management. The data provider and data consumer only illustrate the components and interfaces used for their role. Further, Tab. 3 provides further information on how the design objectives are fulfilled by the architecture. In the following, the final architecture, that is an adaptation of the already available basic Catena-X architecture [Ca25], will be explained per component.

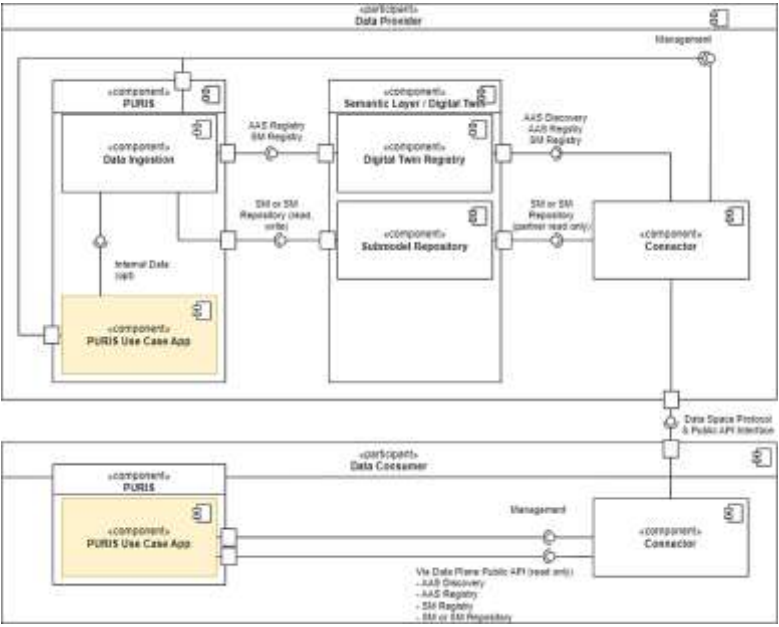


Fig. 2: The final reference architecture as a UML component diagram (after Design Iteration 2)

The **Connector** component is used in a data space to share data assets securely. It allows data providers to manage data assets, their meta data and their terms of use [Gi24]. It allows data consumers to query this information to contract and consume the offered data assets [Ot22b]. The Connector implements data sovereignty by providing access and usage conditions defined by the data provider and must be agreed to by the data consumer [Gi24].

The **Digital Twin Registry** [Ca24g] encapsulates the Asset Administration Shell (AAS) specification's Registry and Discovery interfaces. These use shell descriptors and submodel descriptors to provide meta information about the digital twins and their aspect-related data. The specific asset IDs make a twin discoverable. The submodel descriptor contains a reference to the submodel's semantic description submodel and its endpoint information at the Connector to directly access the information. The data provider registers

the digital twins after creating the data assets in the Connector. The Digital Twin Registry and its APIs are registered as a data asset. The data consumer uses the Discovery and Registry interfaces through the Connector to find the materials and their information.

The **Submodel Repository** provides the aspect-related information of the digital twins. The data provider exposes the read interface for the JSON-only serialization of a submodel via the Connector. Therefore, the submodel is registered as a data asset in the Connector. As mentioned before, the linkage must be defined correctly in the submodel descriptor [Ca24g]. The data provider may use different approaches to provide data via that interface, for example, it updates the repository using the standardized APIs whenever data changes.

The **PURIS Use Case App** implements the use case using internal data. For the supply reliability, such an application matches and compares its supply chain-related information with the partner's corresponding information. Besides visualizing information, the component can provide monitoring and alarming capabilities. These are recommended to identify materials of critical supply. To get partner information, it uses the AAS Discovery and Registry Interface. Using these interfaces, it identifies the digital twins, their data and the corresponding data assets. It accesses the data via the Submodel or Submodel Repository interface [Ca24g]. When handling a partner's information, the application must follow the access and usage conditions defined by this partner within the Connector's contract definitions. To get partner information, the PURIS component uses a data provider's interfaces through the Connector.

The **Data Ingestion** component provides internal data following the PURIS definitions to both the PURIS Use Case App and partners in the data space via the Connector. To achieve the second, it transforms the internal data into the target semantics and updates the submodels in the Submodel Repository component. When the data provider sets up the data assets and contracts in the Connector, this component takes care of creating the offerings in the Connector while correctly registering the digital twins with their submodels in the Digital Twin Registry to make them discoverable.

5 Demonstrating the reference architecture

The architecture has been evaluated and demonstrated across two episodes. After an initial ex-ante evaluation following [Pe12], an ex-post evaluation has been conducted in the second design iteration. The evaluations have been performed by instantiating a reference implementation that has been published as free open-source software [Co24b]. Within the instantiation, the authors implemented the technical basis and stock information exchange. Further information exchanged and the major parts of the frontend (see Fig. 3) have been implemented by other team members. This section covers the instantiation and the evaluation within Design Iteration 2 and describes how the DOs are instantiated.

The reference architecture has been implemented within Catena-X, an automotive data space, based on its enterprise architecture [Ca25]. We have reused the information

exchanged by Opriel et al. [Op24], extended it with delivery information and redesigned it to suit more domains. To exchange information in the Catena-X data space, the Eclipse Tractus-X EDC [Co25b] has been used. In Catena-X, the usage conditions are defined using agreements, purposes and references to actual contracts [Co24c]. The Connector verifies technically that a partner has signed the Data Exchange Governance agreement. The reference application only contracts offerings with a purpose meant for PURIS.

The information related to the digital twin has been modelled using the semantic aspect meta model [Co25a]. Following the given anti-trust regulations, the information has been designed so that all information must be allocated to the partner. To lower the resources needed, the reference implementation covered the PURIS Use Case App and parts of the data ingestion component monolithically. A local deployment has been created using Docker Compose. It provides the reference implementation, a Connector [Co25b] and a Digital Twin Registry [Co24d] for two participants. The deployment contains other necessary components, such as databases, a wallet mock and identity providers. For demonstration purposes, own and partner-related information has been visualized in a frontend (see Fig. 3). The frontend offers users to enter the partner-specific information for stocks, demands, planned production and deliveries manually, based on their role in a partner relationship. Data ingestion for the operational information and master data, like partner and material information, may use the REST APIs also used by the frontend. During the implementation, the integration always started with stock information and has been extended to span additional information implemented by other project members.

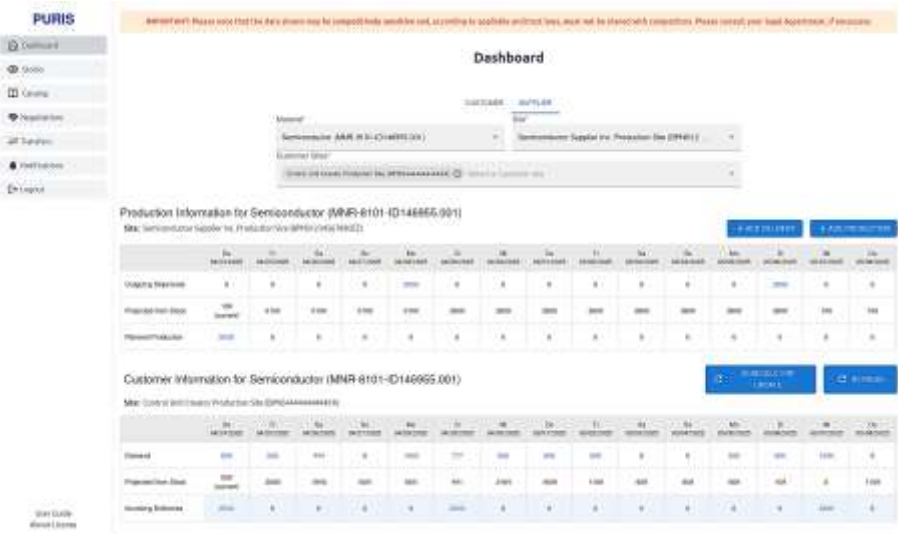


Fig. 3: PURIS FOSS 2.1.0 Material Detail Dashboard from the supplier's point of view

All digital twin-related components and interfaces follow the Asset Administration Shell specification in version 3.0 [In23]. The minimum profile for data provisioning, required

by Catena-X, has been implemented [Ca24g]. Due to resource constraints, the Submodel interface has been implemented in the backend of the reference implementation. The data ingestion's responsibility for the registration and discovery process for digital twins has been automated in the backend. A shared "part type" twin, which in Catena-X is defined as a catalog item after sourcing, is used as a digital twin. The necessary identification data, used to look up the digital twins in a partner's Digital Twin Registry, is based on material numbers [Ca24b]. Each partner provides their part of the information to the digital twin. To complete the view, one pulls the partner's part of the information. To enable the information exchange, the reference application creates the data assets on startup. When a partner is created with a role, the backend performs the registration as defined for the Data Ingestion component (see Section 4.2). The information exchange has been demonstrated in a system demo within the SAFe organization using the reference implementation and needed components twice, once for the data consumer and data provider. The reference implementation has been released in the Eclipse Tractus-X [Co24b].

6 Discussion and conclusion

The design objectives have been used to build and evaluate a reference architecture (see Section 4) for supply chain information exchange based on digital twins using a data space. This architecture allows supply chain practitioners to standardize short-term supply chain information exchange with their partners. As the architecture allows for adding further information, practitioners and researchers may investigate and implement information exchange for further use cases. This lowers transaction costs and makes use of the cost exploitation capability of data ecosystems.

The final architecture can be understood as a use case-specific view on the basic Catena-X architecture [Ca25], which has been designed by other teams, for short-term supply chain information exchange. The architecture has been proven to be applicable. In the following, we discuss how the derived DOs are fulfilled to reflect on possible limitations and demands for future research.

DO1 and DO2 have been covered in the information design. All Information shared must be allocated to partners. The information provisioning has been demonstrated by a project partner who derived stock information from a machine execution system's production orders in a single sourcing scenario within design iteration 1. As the stock information definition did not change significantly in the final version, the feasibility has been proven.

Within the project scope, the interoperability (DO3) has been proven for the Design Iteration 1 architecture based on two implementations. For the digital twin-based architecture with additional information, the reference implementation has been used twice. Nevertheless, the reference implementation was able to prove that a material and information-specific exchange based on digital twins is feasible (D04). The domain unspecificness of information has not yet been proven (DO5.1). For lead times longer than four weeks, such as in steel manufacturing, the information has not been designed.

However, the applicability to multiple use cases has been proven as the dashboard (see Fig. 3) represents a view like shortage spreadsheets and includes a stock projection with a visual indicator to monitor upcoming stockouts.

By releasing the reference application as free open-source software and further documentation, we laid the foundation of scalability (DO6.1 and DO6.2). Nevertheless, the interchangeability of components has not been fully considered (DO6.3) as the Submodel interface has been implemented in the backend instead of using a reusable component. Finally, the architecture follows the Catena-X framework [Ca24d] with usage policies referring to signed agreements, contract reference, and usage purposes (DO7).

In future research, the automatic allocation in multiple sourcing scenarios should be handled to improve the scalability while remaining compliant (DO1). Moreover, the integration of the other information objects, such as the latest information, should be validated further (DO2). As the reference implementation relies on Catena-X, it should be verified whether it is interoperable (DO3) with other data spaces to build overlapping ecosystems. Further data sovereignty may improve technically by, for instance, exchanging aggregated information instead of actual quantities (DO7).

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Appendix

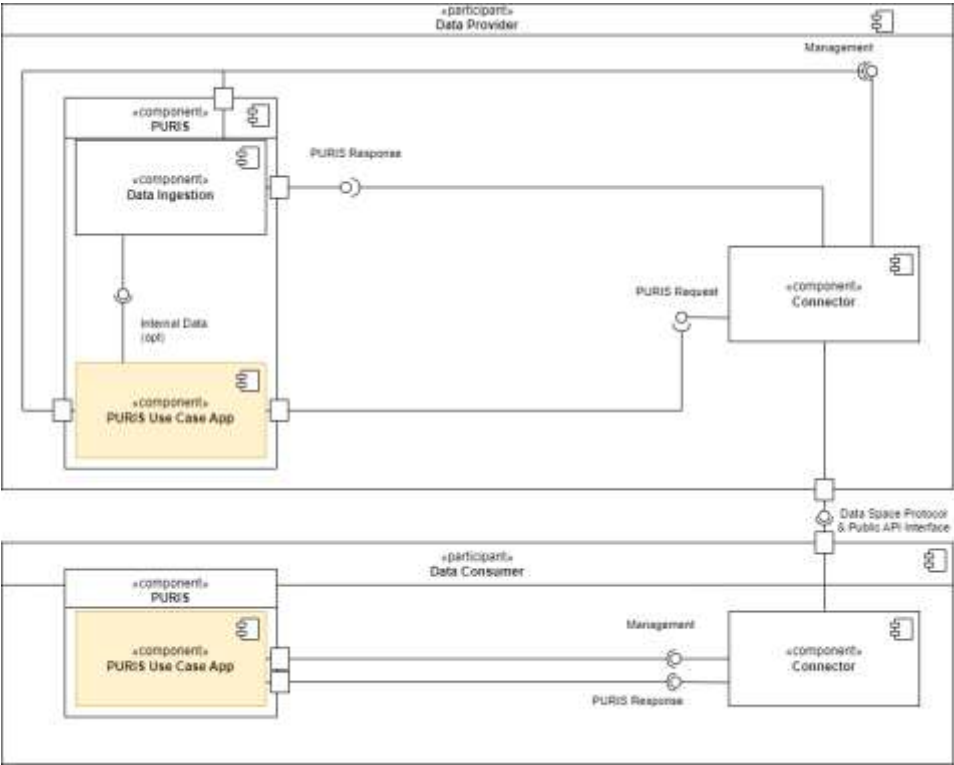


Fig. 4: Architecture after Design Iteration 1 (UML component diagram [Ob17])

Document	Document Type
CX-0118: Delivery Information Exchange 2.0.0 [Ca24e]	Standard
CX-0120: Short-Term Material Demand Exchange 2.0.0 [Ca24e]	Standard
CX-0121: Planned Production Output Exchange 2.0.0 [Ca24a]	Standard
CX-0122: Item Stock Exchange 2.0.0 [Ca24f]	Standard
PURIS KIT 0.3.0 [Co24a]	Adopter Documentation
PURIS FOSS 2.1.0 documentation (architecture, user guide, admin guide) [Co24b]	Software / Adopter Documentation

Tab. 1: Summary of the final documentation available

Design Objective	Derived from
DO1: Compliance by Design	Required by the consortia and laws as explained in section 4.1.
DO2: Up-to-date information	Derived from discussion with practitioners and team members. Updated information must be provided at least daily (e.g. verify supply situation) or multiple times a day (e.g. resolve bottleneck). This is in line with the design requirements "R21 Real-time data exchange" and "R23 Up to-date and correct data" [Op24].
DO3: Follow the same protocols and semantic definition	Required by the consortia. Standardization of the information and its exchange process allows ecosystem participants to switch applications and prevent vendor lock-ins. This is in line with the design requirements "R26 Standardized and structured formats" and "R35 Standardized API" [Op24]
DO4.1: Information exchanged on per-material level	Required by practitioners and consortia team members. Information in supply chains is shared per material, as the information flow follows the contractual relationship and the context of information exchange (see DO5.2)
DO4.2: Easily add further information	Derived from consortia and working sessions with team members. Beside a basic information set, participants may align on additional information to for their "Quid pro Quo" information sharing [Op24].
DO5.1: Domain unspecificness of information	Derived from discussions with the team members on scaling in supply networks. The factory outfitters represented small and medium sized enterprises who often produce make-to-order and participate in supply chains belonging to different domains on the later tiers.
DO5.2: Multiple use cases of information	Derived from experiences of team members to provide multiple participants a potential benefit of information sharing; validated in external feedback workshops. Trust can be built during shortages, to start collaborative usage of the same information. Smaller companies may track their orders while bigger companies monitor their supply.
DO6.1: Reference Implementation	Emphasized by the consortia and acknowledged with team members to provide a starting point for participation. Additionally, using open-source software and packages is associated with building trust [Op24].
DO6.2: Free open-source documentation	Required by the consortia and acknowledged by the team members to provide user-friendlier and openly accessible documentation for adopters
DO7: Preserve data sovereignty	Derived from literature, consortia requirement and confirmed during external evaluation workshops. The

information exchanged in the PURIS case is sensitive. Companies are afraid of misuse. This is in line with the design requirements “R31 Prevent data misuse” and “R36 Combine usage policies and payload” [Op24].

Tab. 2: Data sources of design objectives

Design Objective	Addressed by Component(s)	Explanation
DO1: Compliance by Design	• Data Ingestion • Digital Twin Registry • Submodel Repository	The Data Ingestion component provides information allocated to a specific partner. The Digital Twin Registry and Submodel Repository use access control accordingly.
DO2: Up-to-date information	Data Ingestion	The Data Ingestion must provide the latest information. This can be achieved by e.g., cronjobs or streaming updates of information.
DO3: Follow the same protocols and semantic definition	• Connector • Digital Twin Registry • Submodel Repository	The workflows are handled via the same Connectors Management, Asset Administration Shell and Submodel interfaces (see Fig. 2). The Connector allows to discover the data assets and their policies. The Digital Twin Registry allows to discover the materials and directs to the Submodel Repository for the actual information (consumed via Connector).
DO4.1: Information exchanged on per-material level	• Digital Twin Registry • Submodel Repository	The Digital Twin Registry allows the discovery on per material level for catalog items. The Submodel Repository allows to access specific submodels of these digital twins.
DO4.2: Easily add further information	• Digital Twin Registry • Submodel Repository • Data Ingestion	Further information needs to be semantically described. The Data Ingestion must then provide the new submodel to the Submodel Repository and register it in the Digital Twin Registry.
DO5.1: Domain unspecificness of information	• Submodel Repository (semantic definitions) • Data Ingestion	This DO can only be mapped loosely to the architecture components as the Data Ingestion applies semantic definitions to add

DO5.2: Multiple use cases of information	• Submodel Repository (semantic definitions) • Data Ingestion	the information to the Submodel Repository. Same as DO5.1.
DO6.1: Reference Implementation	• PURIS Use Case App • Data Ingestion	The design objective can't be mapped to the architecture itself but to the PURIS Use Case App and Data Ingestion to evaluate the feasibility of the architecture.
DO6.2: Free open-source documentation	None	The adoption documentation of the architecture and associated reference implementation can't be mapped to the architecture but needs to describe it.
DO7: Preserve data sovereignty	• Connector • PURIS Use Case App • Digital Twin Registry • Submodel Repository	The Connector allows participants to negotiate and evaluate usage conditions of information. Data consumers need to implement policy enforcement (see e.g. [Zr19] for the distribution of these) in their consuming (PURS Use Case App) and providing applications (Digital Twin Registry, Submodel Repository).

Tab. 3: Mapping design objectives to the components